

Graphing Derivatives

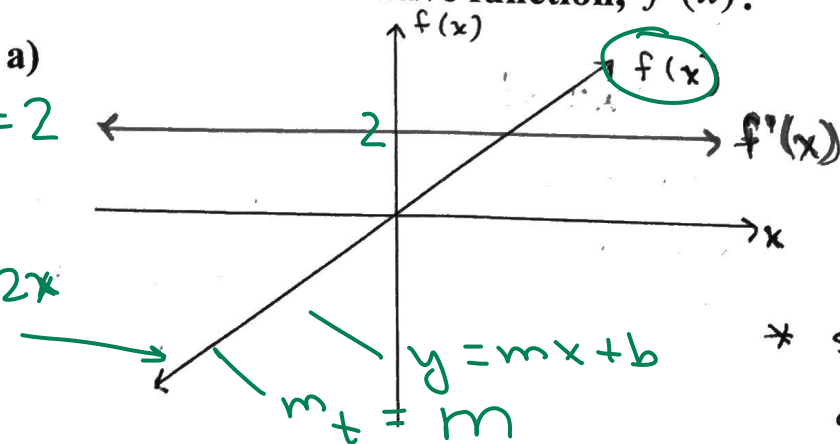
We can relate the graph of a function with its derivative. To do so, remember:

- i) The derivative of a function represents the slope of the tangent line of the function at any defined point, x .
- ii) For polynomial functions, each time we differentiate we reduce the degree of the polynomial by 1.

Notes:

1. If the slope of the tangent is negative at point $x = a$ (ie: $f'(x) < 0$), then the derivative graph is below the x -axis.
2. If the slope of the tangent is positive at point $x = a$ (ie: $f'(x) > 0$), then the derivative graph is above the x -axis.
3. If the slope of the tangent is zero at point $x = a$ (ie: $f'(x) = 0$), then the derivative graph is on the x -axis.

Example 1: For each of the following graphs of $f(x)$ sketch the derivative function, $f'(x)$.

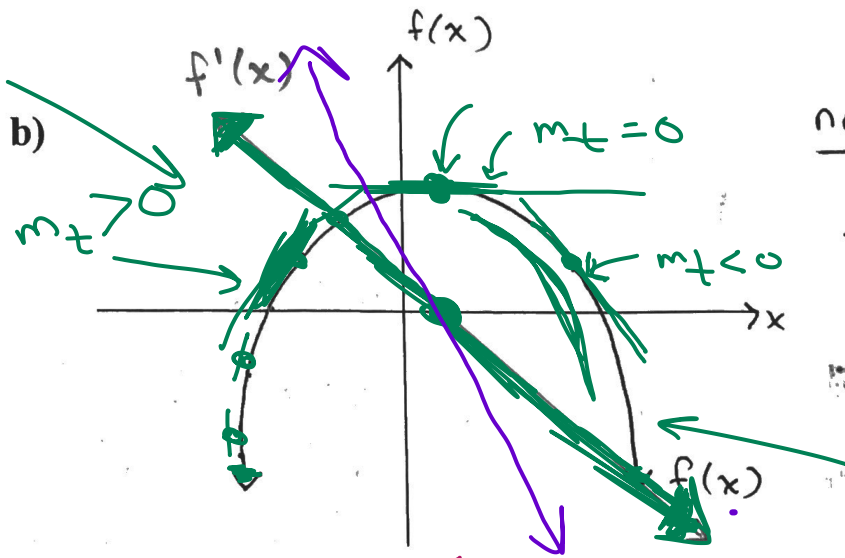


notes:

$f(x)$ is linear

$f'(x)$ is constant

* slope is constant at every point.

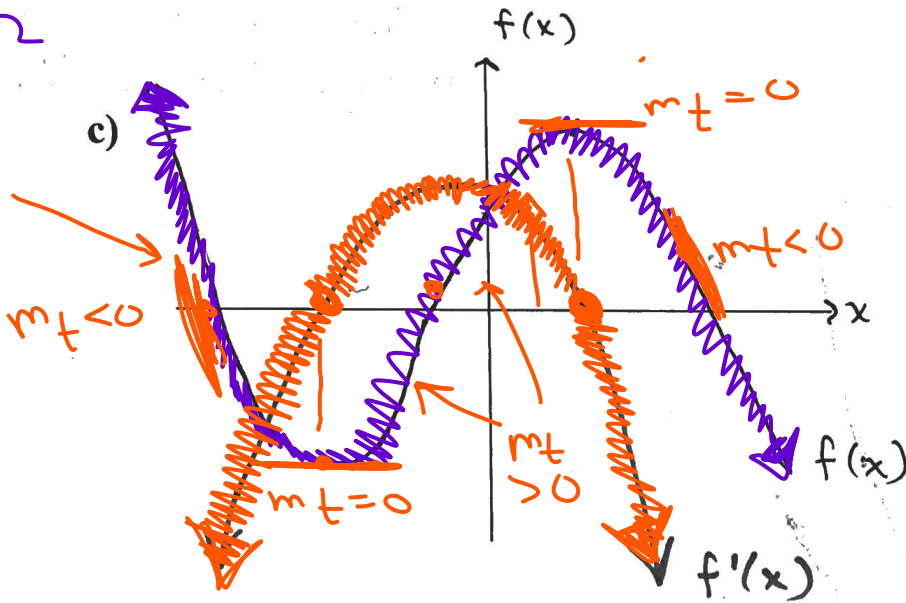


notes:

$f(x)$ is quadratic

$f'(x)$ is linear

Q2



notes:

$f(x)$ is cubic

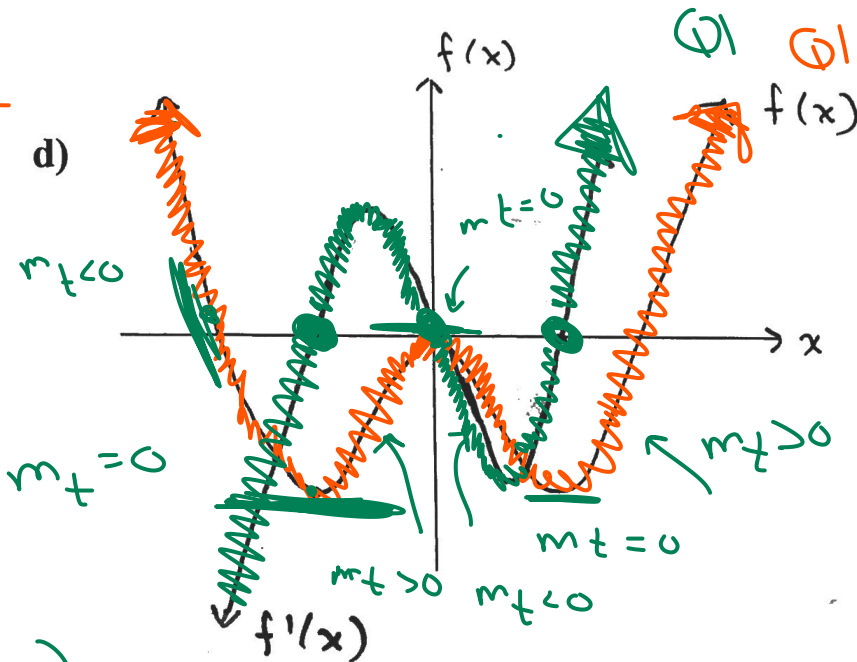
$f'(x)$ is quadratic

L.C. = -ve

$\Downarrow$

④ L.C. $f'(x)$ -ve.

Q2



notes:

$f(x)$ is quartic

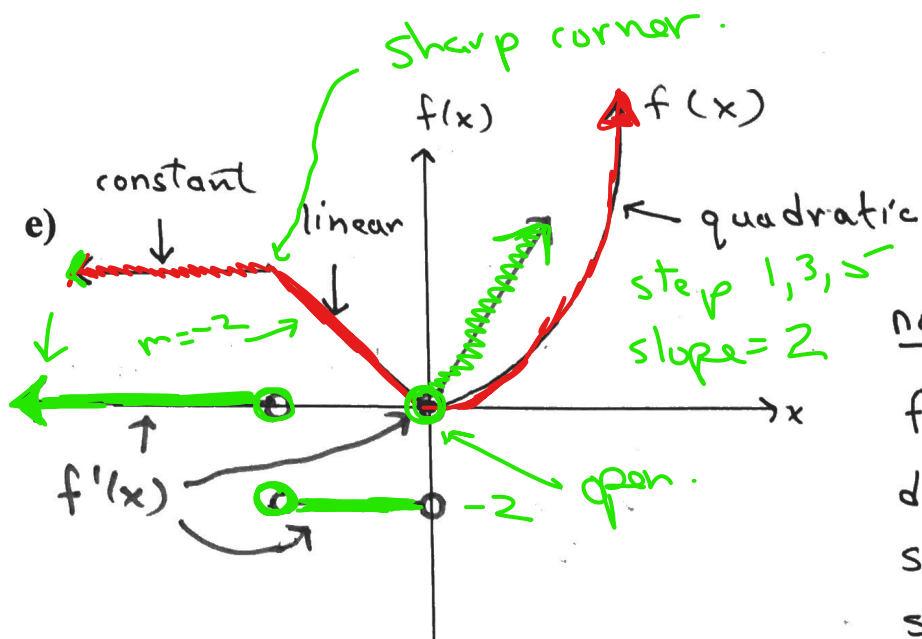
$f'(x)$ is cubic

L.C. = +ve

L.C. $f'(x)$ +ve.



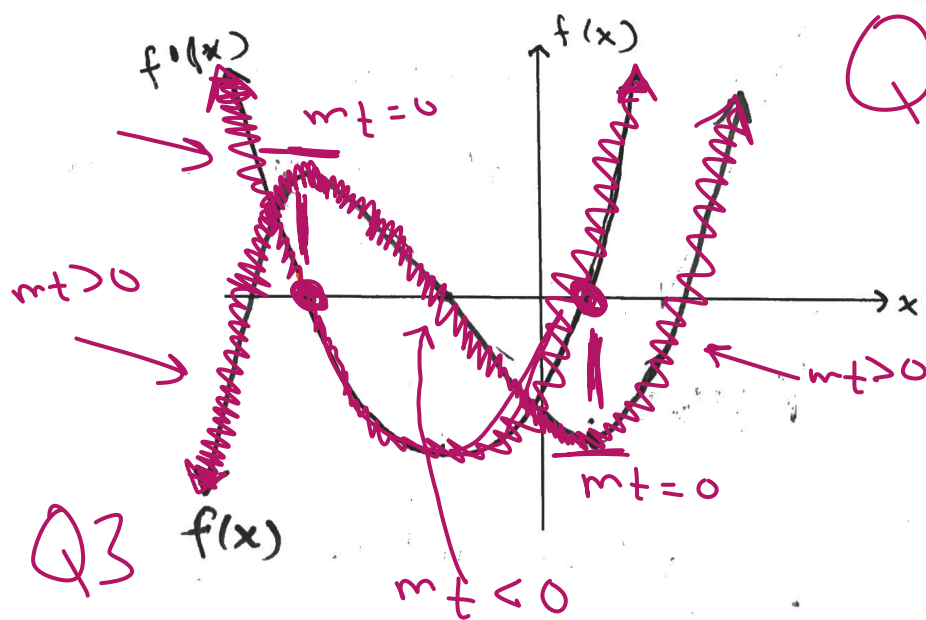
Q3



notes:

$f'(x)$ is not defined at sharp corners since there is no unique tangent line.

Example 2: Given $f'(x)$ below, sketch a possible curve for $f(x)$.



NOT on test!